Jeremy Burke

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Professional Summary

Applied AI & software engineer delivering SBIR/STTR defense and aerospace R&D programs, with experience in computer vision, sensor fusion, simulation, LLM tooling, and technical automation. Builds practical software systems that connect AI models, engineering data, and usable interfaces across $100K–$1M+ efforts.

Experience

Computer Engineer — Beacon Industries — Newington, CT Aug 2025 – Present

- Led SBIR/STTR programs (DoD, NASA, MDA), delivering technical solutions and reports valued at $30K–$250K+ independently
- Developed and trained computer vision models and synthetic sensor pipelines for perception and fusion validation
- Engineered custom plugins for thermal and multi-spectral sensor simulation from raw data streams
- Built AI tools using AWS (S3, EC2), Weaviate, and OpenAI APIs for internal automation and decision support
- Presented technical work in reports and commercialization opportunities directly to government stakeholders

Selected Projects

Technical Paper AI Search Assistant (Public Portfolio Project)

- Built a local RAG-style technical paper assistant for public PDFs using Next.js, FastAPI, Chroma, PyMuPDF, BM25 hybrid retrieval, sentence-transformer embeddings, and Ollama
- Implemented PDF upload, automatic local index rebuild, vector/BM25 hybrid search, and local LLM answer synthesis with document/page-level source traceability

Multi-Spectral Passive Object Detection (Navy SBIR)

- Built ML-based object detection pipeline using synthetic + real multi-spectral data
- Developed 3D physics simulations and sensor plugins to model environmental effects and enable sensor fusion testing

Hypersonic Kill Vehicle Research (MDA SBIR)

- Performed ANSYS simulations to evaluate high-speed system performance and design tradeoffs

Additive Manufacturing R&D (SBIR Phase II)

- Delivered $500K in technical milestones and coordinated external research collaboration

Reverse Engineering (Defense System)

- Leading vendor coordination and technical analysis for legacy electromechanical system reconstruction

Education

University of Connecticut — Storrs, CT Aug 2019 – Dec 2023
B.S. Computer Science & Engineering, Minor Mathematics GPA: 3.3

Technical Skills

Programming: Python, TypeScript, JavaScript, C, C++, Java, SQL
AI/ML: Computer Vision, YOLOv8, RAG, Sentence Transformers, Multimodal Fusion, LLM Systems
Web/Backend: Next.js, FastAPI, REST APIs, Chroma, BM25, Weaviate
Simulation: Gazebo, ANSYS Workbench, Physics-Based Modeling
Cloud/Tools: AWS (S3, EC2), OpenAI API, Ollama, Linux, Git, MATLAB, LaTeX